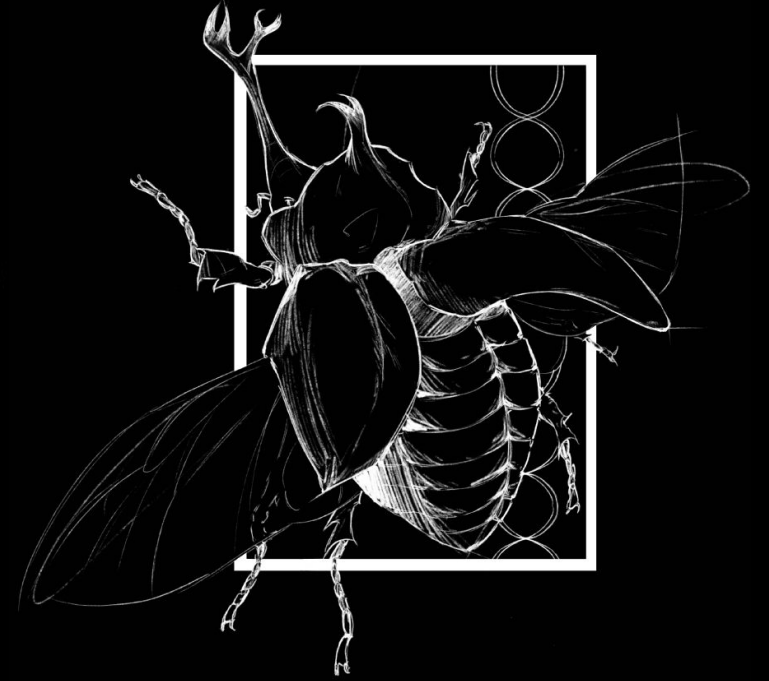




Background

What ecological conditions drive rapid changes in genome structure across animal lineages? Chromosomal rearrangements-such as inversions and heterochromatin expansions-can have profound evolutionary consequences, but their rates of fixation are thought to be shaped by the balance of mutation, selection, and genetic drift. The Drift Barrier Hypothesis predicts that in small, isolated populations, genetic drift can overpower selection, allowing even mildly deleterious chromosomal changes to persist and become fixed. Although this model predicts that ecological fragmentation should accelerate chromosomal evolution, direct evidence from wild mammals is sparse. Cetaceans provide an exceptional opportunity to test these ideas, as their lineages occupy both highly connected pelagic, or open-ocean, environments and fragmented coastal or riverine habitats-contexts likely to produce stark contrasts in population structure and effective population size. By applying phylogenetic comparative methods to genome and habitat data from roughly 18 cetaceans species, **this study tests whether ecological fragmentation is associated with higher rates of repetitive element accumulation.** This approach promises to clarify how population size shape the tempo of genome change.

Study Design & AI

In this study, I investigated whether cetacean species living in fragmented (coastal/riverine) habitats exhibit higher rates of repetitive element accumulation than those in stable pelagic environments. I compiled genomic data (percent repeats) and ecological information (habitat types) for 19 cetacean species with high-quality genome assemblies, drawing from published species accounts, conservation databases such as the IUCN Red List, and genomic sources including the NCBI Assembly Database and *Frontiers in Marine Science*. Habitat type for each species was categorized as pelagic, coastal/riverine, or both. Percent repeats reflects the proportion of the genome made up of repetitive DNA-sequences repeated many times within the genome-which was found for each species alongside genome size, total masked bases (Mb), scaffold N50, and accession ID.

Large language models including chat.tamu.ai and Claude.ai supported the code development and data interpretation in RStudio, enabling phylogenetic mapping and figure generation. Trait and genomic information were aligned to a time-calibrated cetacean phylogeny, and phylogenetic ANOVA was used to account for shared ancestry, enabling reliable and valid evolutionary inference.

Model Fitting

I used phylogenetic ANOVA (phylANOVA in R) to test whether repeat content differs between cetacean habitat types. This method is important because it recognizes that species aren’t completely independent-closely related cetaceans may have similar genomes just because they share a common ancestor, not necessarily because of their habitat. If we ignored the evolutionary relationships and treated each species as independent, our results could be misleading. PhylANOVA works by using information from the phylogenetic tree to simulate random trait differences, which creates a baseline for what we’d expect to see if there were no real habitat effects. This way, we can compare our actual findings to the simulated baseline and determine if the differences in repeat content between habitat groups are truly meaningful or just a result of shared ancestry. I ran these analysis in RStudio, making sure each step was reproducible and that the phylogenetic relationships among the species were properly accounted for. Overall, this approach helps us get a clearer and more accurate understanding of how genome traits relate to habitat in cetaceans.

Data and Results

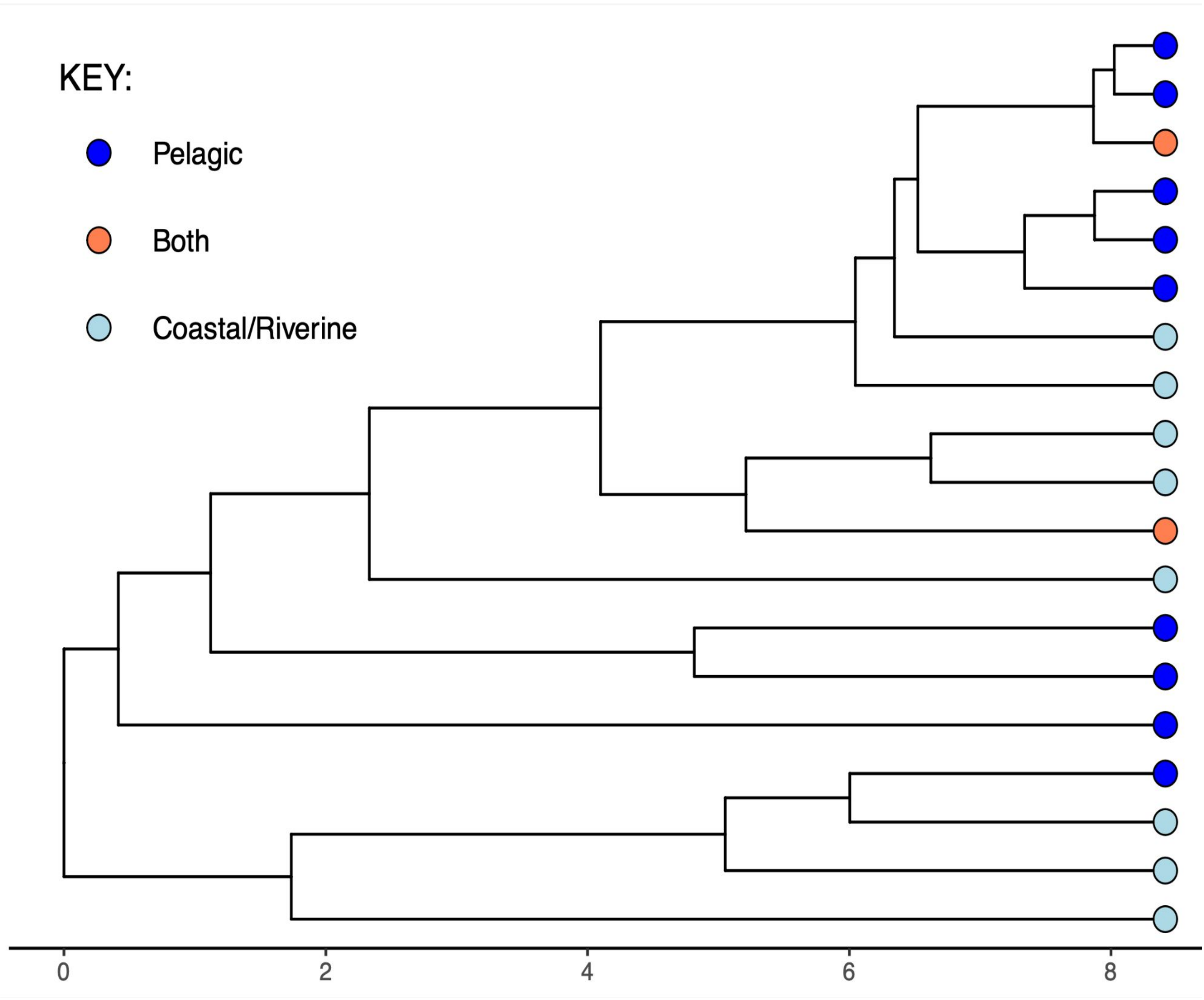


Figure 1. Phylogenetic relationships of 19 cetacean species with habitat type mapped onto the tips. Each tip is colored according to the species’ primary habitat classification: pelagic (blue), coastal/riverine (light blue), or both (orange). This visualization highlights the diversity of habitat use across the cetacean clade.

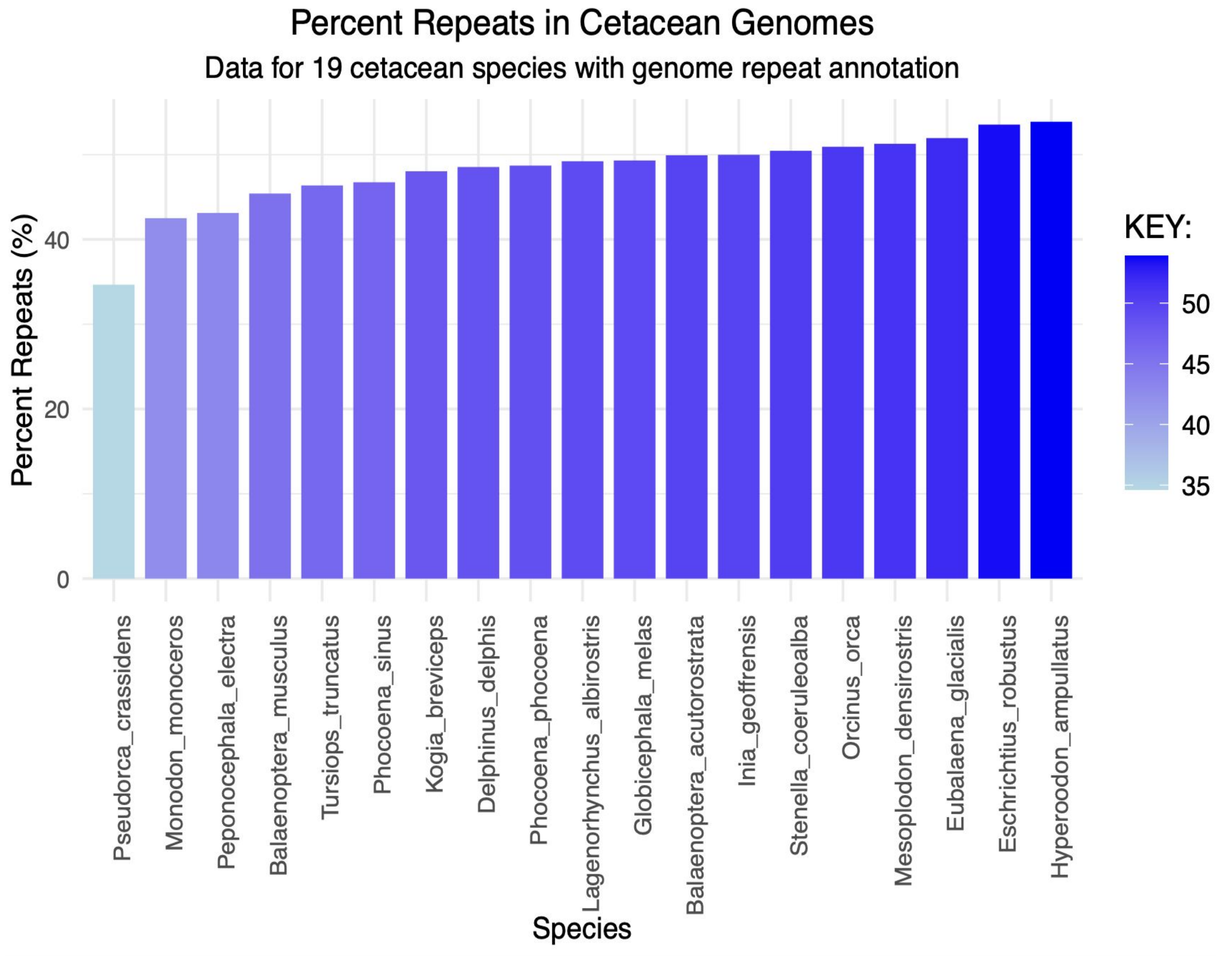


Figure 2. Proportion of repetitive elements in the genomes of 19 cetacean species. Bars are colored by percent repeats, from lower (light blue) to higher (dark blue) values. This data illustrates substantial variation in repeat content among cetacean genomes.

Repeat Content by Habitat Type

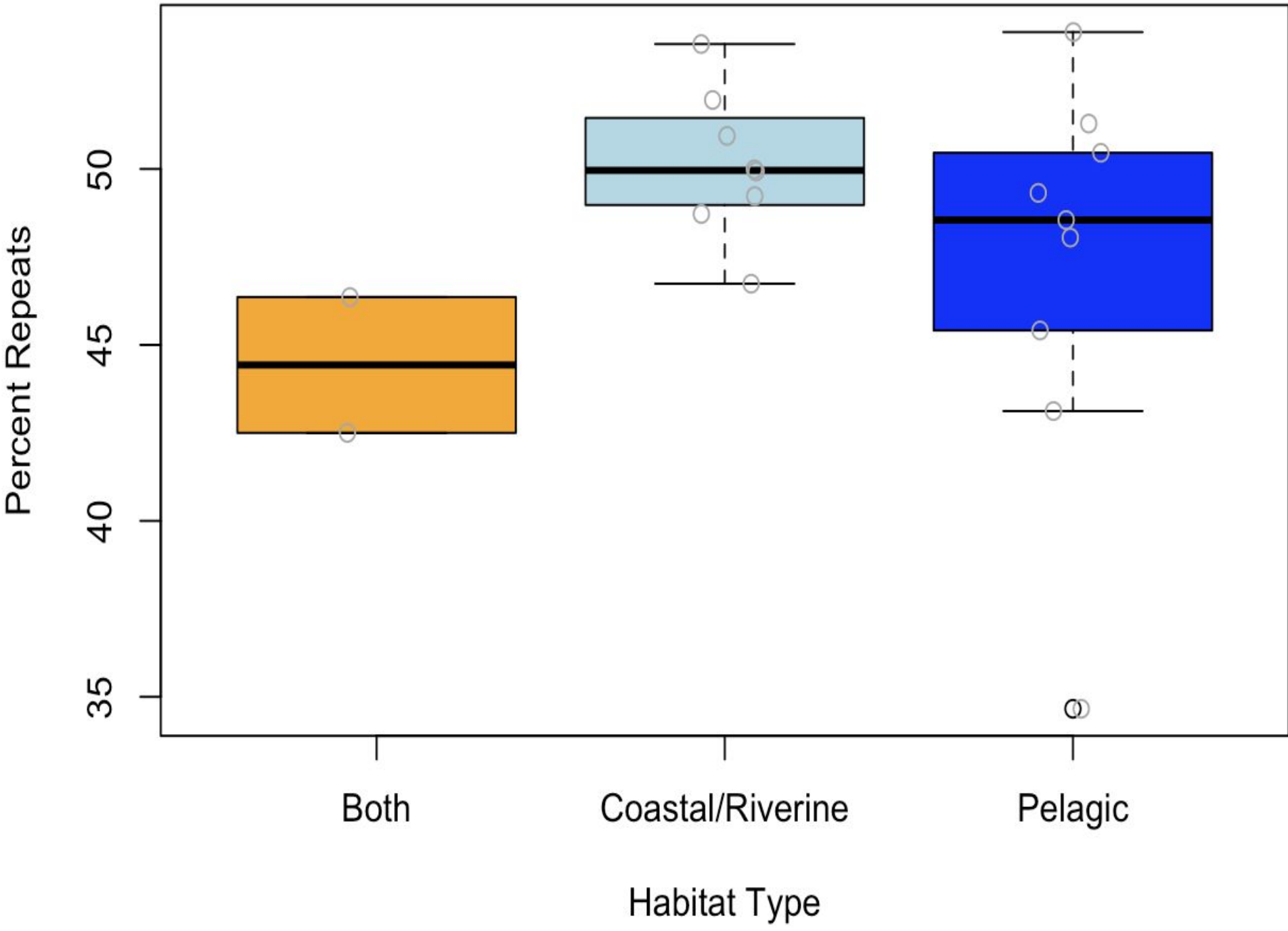


Figure 3. Percent repeat content in cetacean genomes by habitat type. Boxplots show the distribution of repetitive DNA (%) for species classified as Pelagic (blue), Coastal/Riverine (light blue), or Both habitats (orange). Each point represents a single species.

Discussion

My hypothesis was that cetacean species in variable or fragmented habitats (coastal/riverine) would show higher repeat content than pelagic species. Figure 3 suggests coastal/riverine species have higher median repeat content, but the phylogenetic ANOVA was non-significant ($F = 1.83$, $p = 0.169$), and no pairwise comparisons reached significance (all corrected $p > 0.26$). The data do not support habitat type as a significant predictor of repeat content after accounting for phylogeny.

The drift-barrier hypothesis predicts that reduced effective population size in fragmented habitats should facilitate fixation of slightly deleterious mutations, increasing repeat content. While the visual pattern is weakly consistent with this, the statistical results are not. Habitat fragmentation may not be the primary driver of repeat accumulation in cetaceans, or other factors may obscure the signal.

The most obvious limitation is sample size: only 19 of roughly 90 recognized cetacean species had suitable genome assemblies. This reduces statistical power and risks biased representation across lineages and ecological types. Habitat classification presents a second challenge, as many cetaceans use multiple environments or shift habitats seasonally, making clean categorical assignments difficult. Variation in assembly quality and repeat annotation methods across species adds further noise.

Future work should prioritize expanding taxonomic sampling, standardizing repeat annotation pipelines, and replacing broad habitat categories with continuous measures of fragmentation. Greater sampling would improve both representativeness and power, and finer ecological resolution might reveal patterns the current categorical scheme misses.

Conclusions & Acknowledgements

In conclusion, my data does not show a statistically significant difference in mean percent repeat content among cetacean habitat types, after accounting for phylogeny. While I do see visual trends in the boxplot, those differences are not strong enough (i.e., could easily happen by chance) given my data and tree.

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